IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

OPTIMISATION

Thursday, 2 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) :

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OPTIMISATION

1. Consider the (continuous) function

$$f(x,y) = x^4 - x^2 + y^4 - y^2.$$

a) Show that the function is radially unbounded, hence conclude that it has a global minimizer. [2 marks]

b) By using first order necessary conditions of optimality determine all stationary points of the function f.
(Hint: the function has 9 stationary points.) [4 marks]

c) By using second order sufficient conditions of optimality classify the stationary points of the function f determined in part b).
(Hint: the function has 4 minimizers, 1 maximizer, and 4 saddle points.)
[4 marks]

d) Show that all minimizers of f are global minimizers and sketch the level lines of the function f. [4 marks]

e) Consider the Newton's iteration for the minimization of the function f. Note that this has the form (the function N is the same in both equations)

$$x_{k+1} = N(x_k), \qquad y_{k+1} = N(y_k),$$

hence determine the function N. [2 marks]

Consider only the iteration for x. Let $\{x_k\}$ be a sequence which converges to 0, that is $\lim_{k\to\infty} x_k = 0$. Show that the sequence has speed of convergence of order 3.
[2 marks]

Consider again only the iteration for x. Let $\{x_k\}$ be a sequence which converges to $\sqrt{2}/2$, that is $\lim_{k\to\infty} x_k = \sqrt{2}/2$. Show that the sequence has speed of convergence of order 2. [2 marks]

2. Consider the constrained optimization problem

$$\min_{x_1,\, x_2} (x_1 - 1)^2 + \left(x_2 - \frac{5}{2}\right)^2,$$
$$x_1 \geq 0,$$
$$x_2 \geq 0,$$
$$x_1 - 2x_2 + 2 \geq 0,$$
$$-x_1 - 2x_2 + 6 \geq 0,$$
$$-x_1 + 2x_2 + 2 \geq 0.$$

a) Plot on the (x_1, x_2) plane the level lines of the objective function and the admissible set. [4 marks]

b) Show that all points are regular points for the constraints.
(This can be shown graphically or analytically, both give full marks).
[4 marks]

c) Solve, graphically, the considered problem.
(Hint: the optimal solution is at a point in which a level line of the cost is tangent to the boundary of the admissible set.) [4 marks]

d) Write the first order necessary conditions of optimality for the considered problem and show that there exists multipliers such that the optimal solution determined in part c) satisfies such necessary conditions.
(Hint: 4 out of 5 multipliers are zero at the optimal point.) [4 marks]

e) Show that the candidate optimal point determined in part d) satisfies the strict complementarity conditions hence, by applying second order sufficient conditions of optimality, show that such a point is a local minimizer for the considered problem. [4 marks]

3. Consider the optimization problem

$$P: \begin{cases} \min\limits_{x,\,y,\,z} (x+1)^2+(y+1)^2+z^2, \\ x=0, \\ y=0, \end{cases}$$

and the equivalent problem

$$\tilde{P}: \begin{cases} \min\limits_{x,\,y,\,z} (x+1)^2+(y+1)^2+z^2, \\ x^2=0, \\ y^2=0. \end{cases}$$

a) Write first order necessary conditions of optimality for problem P and determine candidate optimal solutions. [4 marks]

b) Write first order necessary conditions of optimality for problem $\tilde{P}$ and explain why these do not return any candidate optimal solution. [4 marks]

c) Write the sequential penalty function F_ε for problem P and determine its global minimizer $p_\varepsilon^\star = (x_\varepsilon^\star, y_\varepsilon^\star, z_\varepsilon^\star)$. Show that $p_\varepsilon^\star$ converges, as $\varepsilon \to 0$, to the solution determined in part a). Use the function F_ε and $p_\varepsilon^\star$ to determine an approximation of the optimal multiplier. [4 marks]

d) Write the exact penalty function G for problem P. Determine its stationary point and show that this coincides with the optimal solution determined in part a). Determine conditions on $\varepsilon > 0$ such that the stationary point of G is indeed a (global) minimizer. [4 marks]

e) Write the sequential penalty function $\tilde{F}_\varepsilon$ for problem $\tilde{P}$ and show that it is globally convex for all $\varepsilon > 0$. Hence argue that it has a global minimizer and that, as $\varepsilon \to 0$, this converges to the optimal solution of problem $\tilde{P}$ (which is the same as the solution of problem P). [4 marks]

4. a) Consider the problem of optimally splitting traffic of rate r over three parallel links. Let x_1, x_2, and x_3 be the flow rates over the links. The link costs, for $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$, are given by the functions

$$c_1(x_1) = (x_1+1)^2 - 1, \qquad c_2(x_2) = 2(x_2+1)^2 - 2, \qquad c_3(x_3) = 3(x_3+1)^2 - 3.$$

Given $r > 0$, the optimal splitting traffic problem is

$$OST : \begin{cases} \min\limits_{x_1,\, x_2,\, x_3} \; c_1(x_1) + c_2(x_2) + c_3(x_3), \\ x_1 + x_2 + x_3 = r, \\ x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0. \end{cases}$$

i) Write first order necessary conditions of optimality for the considered problem. [4 marks]

ii) Determine a value of $r > 0$ such that the optimal solution $(x_1^\star, x_2^\star, x_3^\star)$ is such that $x_1^\star > 0$, $x_2^\star > 0$, and $x_3^\star = 0$.
(Hint: the value of r is not unique, it is sufficient to determine one.)
[4 marks]

b) Consider the problem of computing the cubic roots of -1. These are the complex solutions of the equations

$$z^3 + 1 = 0,$$

that is

$$z_1 = -1, \qquad z_2 = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \qquad z_3 = \frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

The roots can be determined by solving the *optimization* problem

$$Z : \begin{cases} \min\limits_{x,\, y} 1, \\ \mathscr{R}(z^3+1) = 0, \\ \mathscr{I}(z^3+1) = 0, \end{cases}$$

where $z = x + iy$.
(Hint: $\mathscr{R}(z^3+1) = x^3 - 3xy^2 + 1$ and $\mathscr{I}(z^3+1) = 3x^2y - y^3$.)

i) Write first order necessary conditions of optimality for problem Z. [4 marks]

ii) By using the conditions in b.i) determine candidate optimal solutions for problem Z and show that these are indeed the cubic roots of -1. [4 marks]

iii) Suppose one modifies the cost in problem Z to be an arbitrary function $f = f(x,y)$. Suppose that the function f is defined and differentiable for all (x,y). Explain why applying the first order necessary conditions of optimality to the modified problem returns the three cubic roots of -1, regardless of the selection of f. [4 marks]